

# Puffin Drone Industry Application Product DescriptionV2.0 \_8-29-2025

## Puffin Drone Application Product Description V2.0 \_9-30-2025

Welcome to use the Puffin 4G Module for Drone Application. This document focuses on how the Puffin module connects to the drone's flight controller and PC ground station. Please note that Puffin UAV is only compatible with windows PC Mission planner and QGroundControl.

Video link for youtube product and usage instructions:

[Puffin 4G  
Tutorial]<https://www.youtube.com/watch?v=uQ357d5AZ9g&list=PLKhFZhesT3kcvlQKE6AGf0Mk-wf7wze07>

Website: [www.puffinboard.com](http://www.puffinboard.com)

## Overview of PuffinProduct

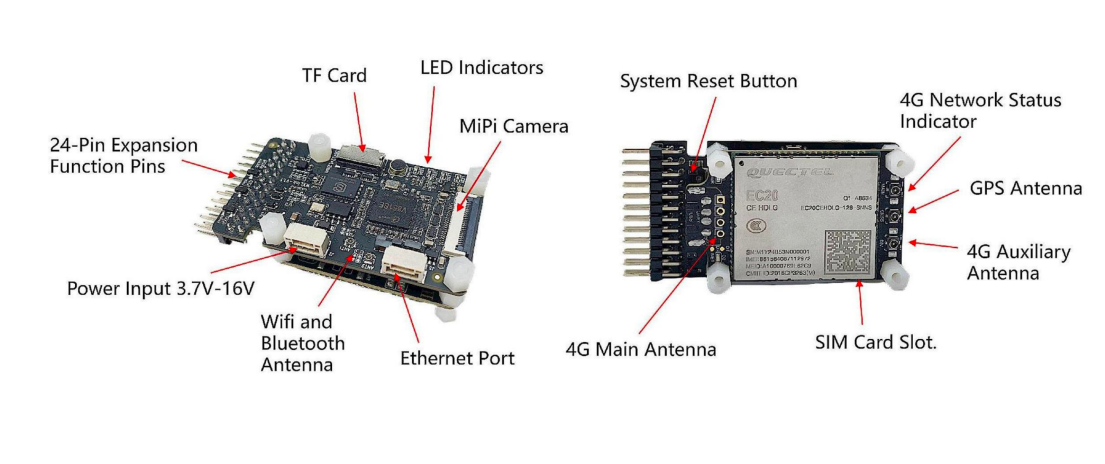
Puffin (aka TEL801) is a lightweight remote control module for unmanned devices (video transmission, data, control). Relying on the 4G cellular network and adopting WebRTC P2P technology, it achieves ultra-low latency video transmission (~100ms RTT) and control (~30ms RTT).

|                                               |                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------|
| <b>CPU</b>                                    | <b>Cortex-A7 core@900MHz</b><br><b>RISC-V@600MHz</b>                      |
| <b>NPU/ DDR</b>                               | On-chip 0.5T NPU<br>On-chip 64M DDR                                       |
| <b>Operating System</b>                       | Tina-Linux                                                                |
| <b>4G LTE Cellular<br/>Network Connection</b> | Cellular Network: 4G LTE MIMO,<br>Upstream: Below 50Mbps<br>Line: 150Mbps |
| <b>Other wireless connections</b>             | WIFI: 2.4G<br>Bluetooth: 2.4G<br>GNSS: GPS/GLONASS                        |

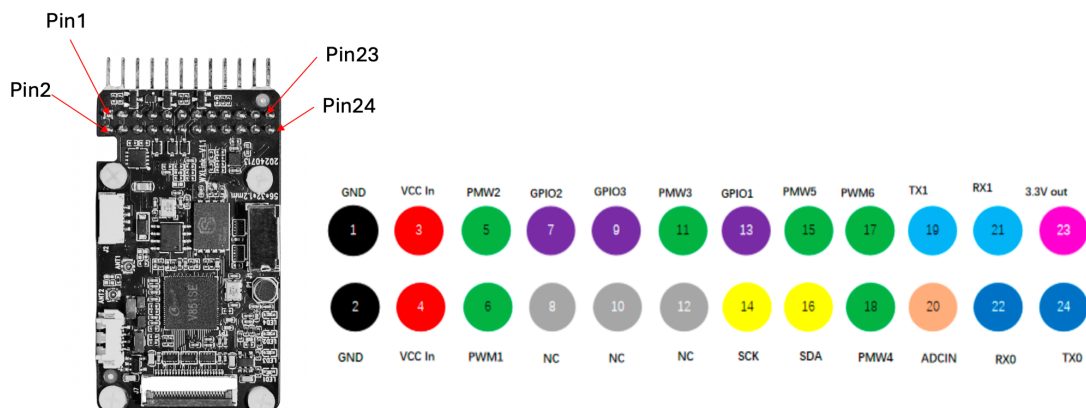
点击图片可查看完整电子表格

# Hardware Interface Description

## 1. Hardware Interface Description



## 2. Serial connection with the drone flight controller

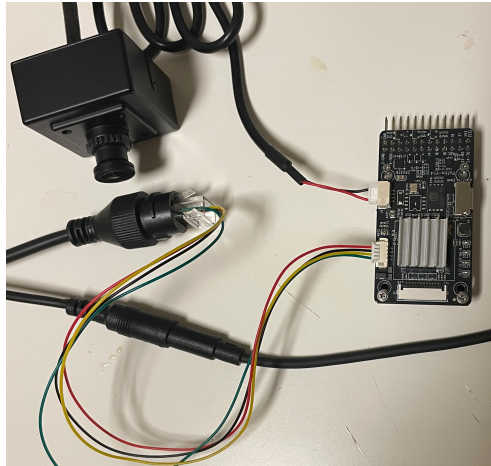


First, you need to familiarize yourself with the 24 PIN pin expansion interface of TEL801. When connecting to the flight controller, you only need to connect GND, TX1 (pin19), and RX1 (21) to the corresponding ports of the flight controller's UART (TELEM). Note that if the flight controller's UART port can output a stable voltage (4-16V), you can connect Vcc to directly power Puffin through the flight controller's UART port. Otherwise, please do not connect the Vcc lead, and Puffin needs to be powered from another power source.

The pins most relevant to our current configuration are as follows:

- GND (Ground): Pin 1, Pin 2
- Vcc in (Power Input): Pin 3, Pin 4
- Serial Port (2 sets):
  - Pins 22 and 24 **Debug** Serial Port
  - Pins 19 and 21 are connected **to the serial port of the flight controller**

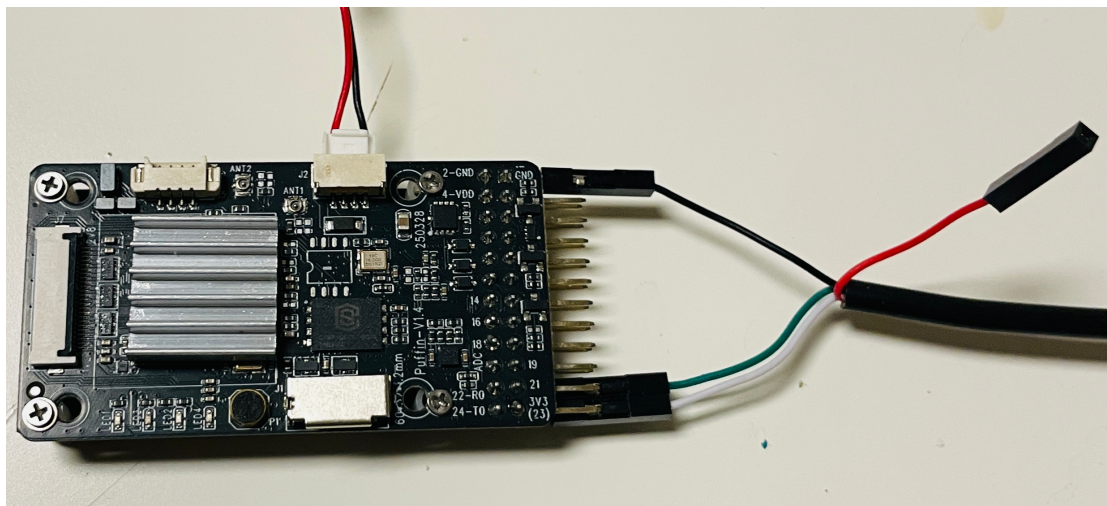
### 3. Connect RTSP Camera (Gimbal)



The network port of Puffin can be connected to the RTSP protocol camera of a drone. The hardware connection method is as follows: Connect the 4-pin network port of Puffinto the RTSP camera with a standard network port using the included 4-pin network port adapter cable, as shown in the figure. The RTSP camera requires separate power supply.

When the Puffinmodule connects to an **RTSP camera, the default IP Address is**: rtsp://192.168.0.154:554/ch01.264. If users need to connect to an RTSP camera with their own address, they need to modify the configuration file inside the Puffinfirmware, It is very important to set the network segment of the RTSP camera to segment 0. The method is as follows:

1. Connect to the P22 and P24 pins of the Puffin24-pin expansion interface using the included serial debugging cable as shown in the figure to debug the serial port. Connect the **white** wire to P24 and the **green** wire to P22. And Black wire to Pin 1 or 2 (GND). You don't need to connect the red wire.



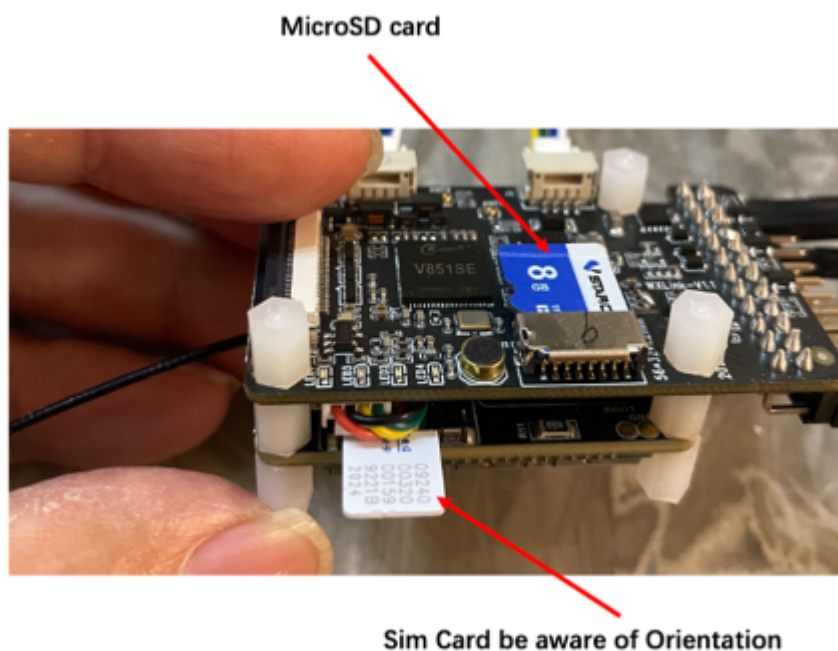
2. Use debugging tool software such as Xshell to view the Linux startup process of TEL801, and after Linux has started, enter commands

cat /usr/local/bin/conf.yaml, change the line rtspURL: rtsp://192.168.0.154:554/ch01.264 to the customer's own RTSP IP Address. **Note that the network segment is 0.**

3. If the customer must retain the network segment of their own RTSP camera, or if the camera cannot connect and no video is output, please contact Puffin's technical support.

## 4. SD Card and SIM Card Insertion

First, insert the SD card into the designated card slot on TEL801. This SD card will be used to store important data and files required for operation. If using the 4G version, please also insert the SIM card into the corresponding card slot. It is recommended to use your own daily mobile data SIM card, **do not use IoT SIM cards! (They may have data limits)** Be sure to note that you must insert and remove the SD card and SIM card while the device is powered off! Otherwise, it may cause permanent damage to the hardware!

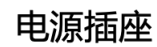


## 5. Power Supply

Puffin supports multiple power supply methods and 4-16V DC voltage.

### 5.1 Interface Power Supply





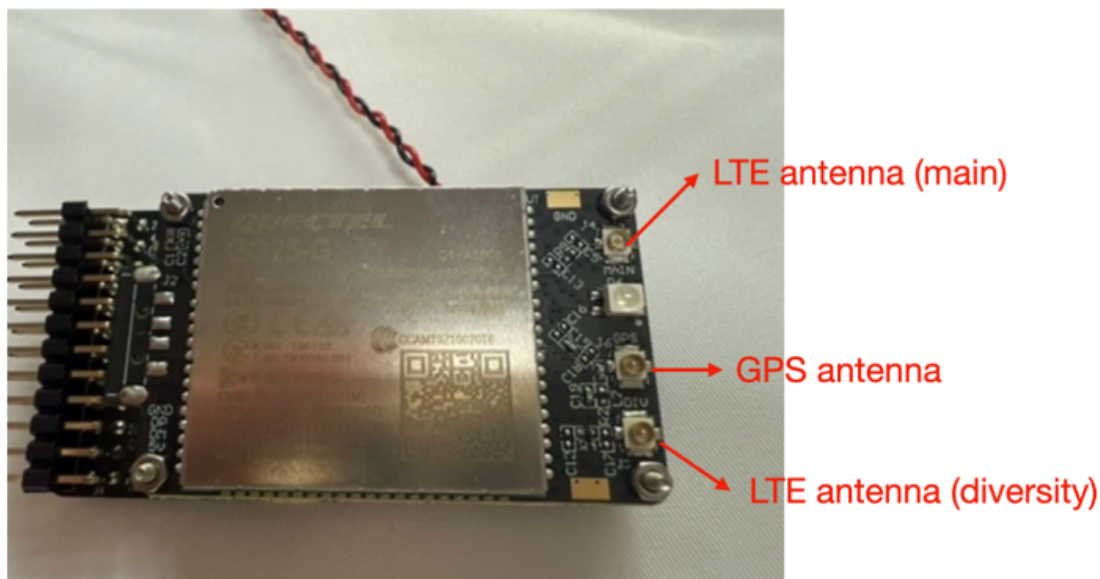
## 5.2 Connect to external lithium battery power supply

A photograph of a Raspberry Pi Zero W board with a USB-C to UART module connected. Red lines point from labels to the connections: GND (black), VCC 4V-16V (red), TX (green), and RX (blue).

## 6. Antenna Installation

## 6.1 LTE/GPS Antenna

The LTE antenna and GPS antenna are located on the lower layer of the laminated board. LTE uses dual MIMO antennas.



## 6.2 Antenna Fixed Position

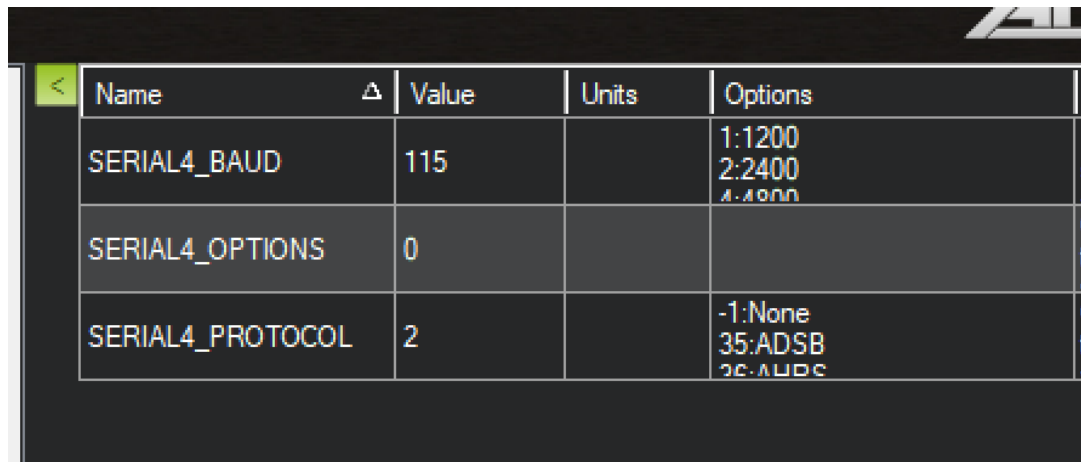
Be sure to pay attention to the installation position of the antenna. The antenna we provide is an FPC antenna. Do not directly attach the antenna to **metal** or **carbon fiber** sheets, as the antenna performance will be greatly affected! The antenna can be fixed on plastic components or suspended. Note that many fuselages are made of carbon fiber, which has a shielding effect, **do not directly fix the antenna to the carbon fiber fuselage!** A plastic foam at least 5 cm thick can be used to isolate the antenna from the fuselage.



## Puffin Connects to Flight Controller

### Flight Control Parameter Settings

First, set the correct flight controller parameters. Connect the flight controller to Mission Planner. Find the corresponding UART port in config->Full parameter list. If using ArduPilot and Serial Port 4 is used, set the parameters in Serial4 as shown in the figure below.



| Name             | Value  | Units | Options                           |
|------------------|--------|-------|-----------------------------------|
| SERIAL4_BAUD     | 115200 |       | 1:1200<br>2:2400<br>3:4800        |
| SERIAL4_OPTIONS  | 0      |       |                                   |
| SERIAL4_PROTOCOL | 2      |       | -1:None<br>35:ADSB<br>36:MAVLink2 |

Baud rate:115200

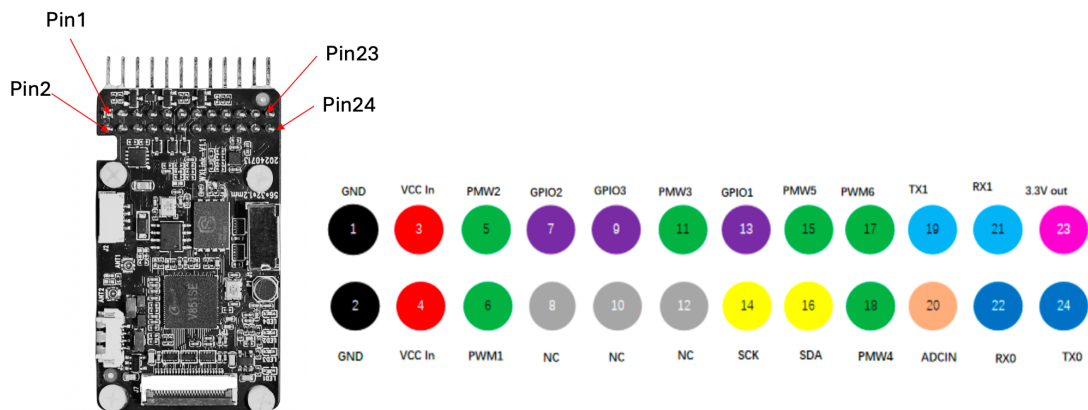
Options:0

Protocol:2

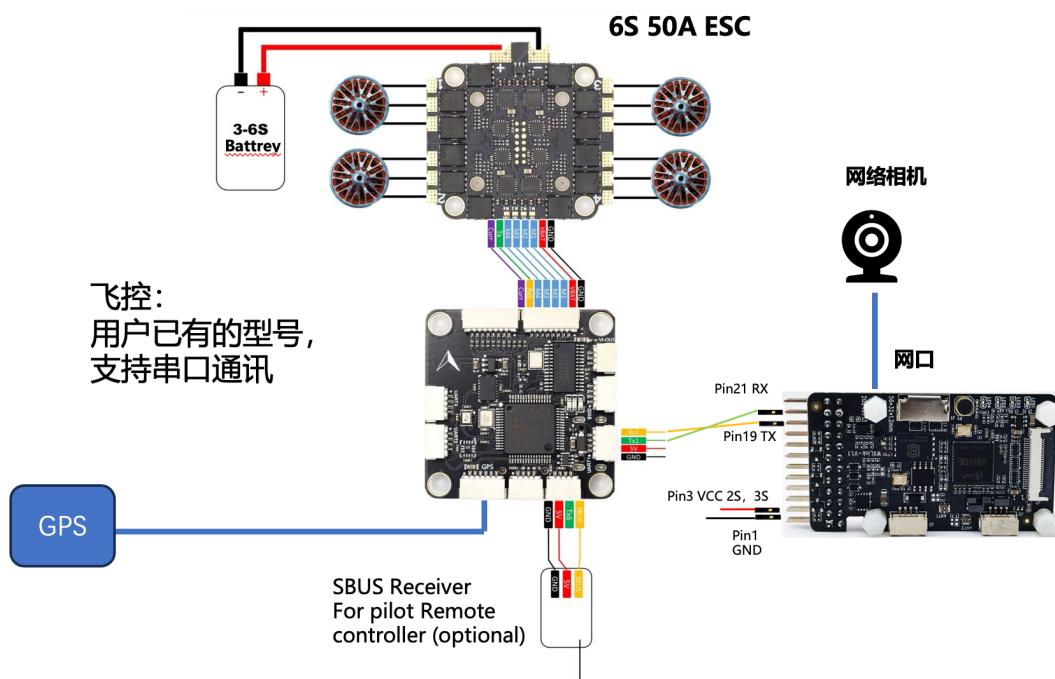
After the settings are completed, click "write params", then power off and power on the flight controller to restart it, and confirm that the parameters are set to the correct values. The same principle applies to PX4 flight controllers. Confirm that the UART port of the flight controller supports the Mavlink2 protocol.

## Connect the flight controller to TEL801

The flight controller is connected to Puffin via serial port, with the principle being serial port pass-through. The flight controller serial port configured in the previous step needs to be connected to TEL801. As shown in the figure below, the TX line of the flight controller UART needs to be connected to the RX (pin21) of TEL801, and the RX of the flight controller is connected to the TX (pin19) of TEL801. **It is necessary to** connect the GND of the flight controller UART to the GND of TEL801. Whether to connect Vcc depends on the power supply situation of TEL801. If the flight controller UART serial port can output a stable voltage, it can be connected and used as the power supply for TEL801. If Puffin already has an external power supply, then it must **not** connect Vcc.

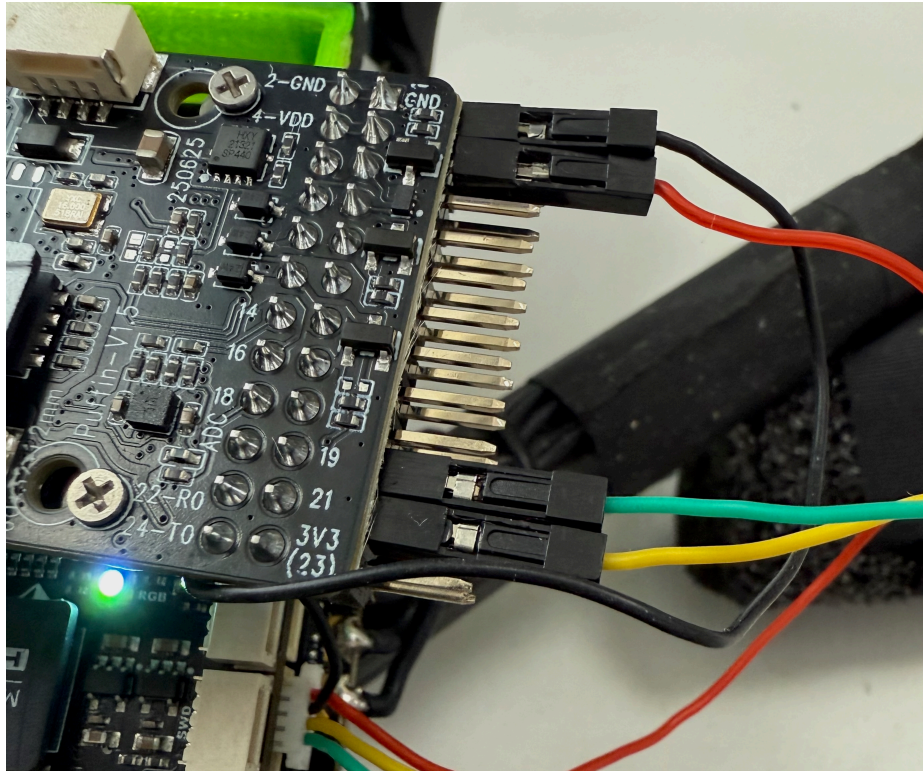


Flight Controller: Existing models owned by the user, supporting serial communication



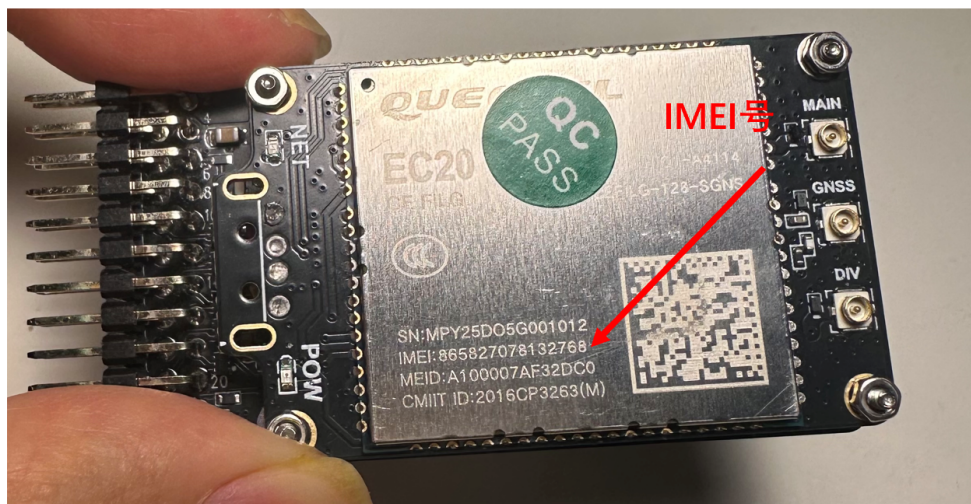
The following figure shows the connection between Puffinand MircoAir743-V2. The serial port of MicroAir743 supports a constant voltage of 5V, which has always been used as the power supply for TEL801.





## Get device IMEI number

The IMEI number of this device can be seen below the metal casing of the 4G module on the back of the module



## Camera Connection:

TEL supports RTSP network cameras, requiring cameras that can change their IP addresses. The network port of Puffin is connected to the network camera. Please set the IP address of the network camera to `rtsp://192.168.0.154:554/ch01.264`, with the encoding format being

H.264, resolution 1080P, and frame rate 30fps. If the camera cannot change its IP settings, please contact customer after-sales service.

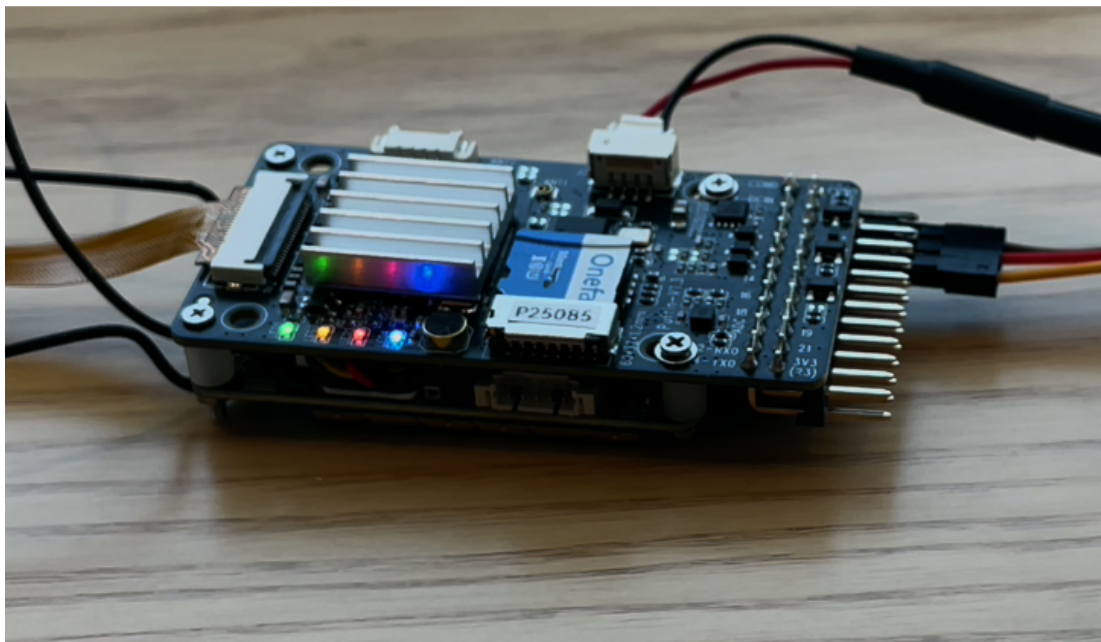
## Check List

Please confirm before powering on

- SIM card inserted correctly, MicroSD card inserted correctly
- 4G Antenna Installed Correctly
- Serial port is correctly connected to the flight controller
- The network port is correctly connected to the pod (if the pod requires external power supply, pay attention to powering it on).

## Power On

Power on TEL801. Puffin can be powered by the 2S to 4S battery of the drone, with the maximum power supply voltage not exceeding 18V. Pay attention to observing the blue LED light in the figure below. A blinking blue light indicates successful network connection. Otherwise, please power off and then power on again.



After the power cord is connected to the power supply for a dozen seconds, ensure that the red LED on the front of the module **remains constantly lit**, and the blue LED **flashes**, indicating that the module has successfully connected to the signaling server. If the blue LED does not flash, please check if the SIM card has data. If it does not flash after power-on, please contact technical support.

Blue Light Uncensored Frequently Asked Questions

- The SIM card has run out of data
- Antenna not installed

- The 4G signal in the indoor environment is very weak

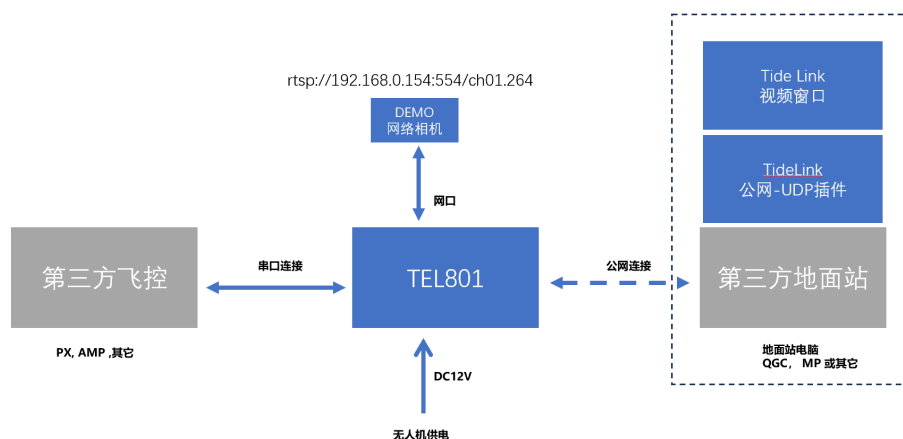
## Puffin connects the flight controller and the ground station

### Principle

Puffin can be easily connected to the flight controller of a drone via a serial port. The block diagram of the overall principle is as follows:

Puffin establishes a public network-based UDP data transmission control channel via WEBRTC between the flight controller (compatible with PX4 and ArduPilot) and the PC ground station (QGC, Mission Planner, or compatible customer ground station). This channel is actually a serial port pass-through channel. In an environment with good network signals, commands from the ground station can be passed through to the flight controller to control the drone, and the drone's flight parameters can also be passed through to the ground station for parsing and display on the ground station interface. Meanwhile, Puffin also establishes an image transmission channel between the pod (RTSP camera or gimbal camera) on the drone and the PC. By launching the Web app included with the software on the PC browser, high-definition real-time images from the drone pod can be displayed, and the RTSP camera with gimbal functionality can also control the gimbal via network protocols.

**TEL801 4G图传控制模块的飞控地面站连接原理框图**



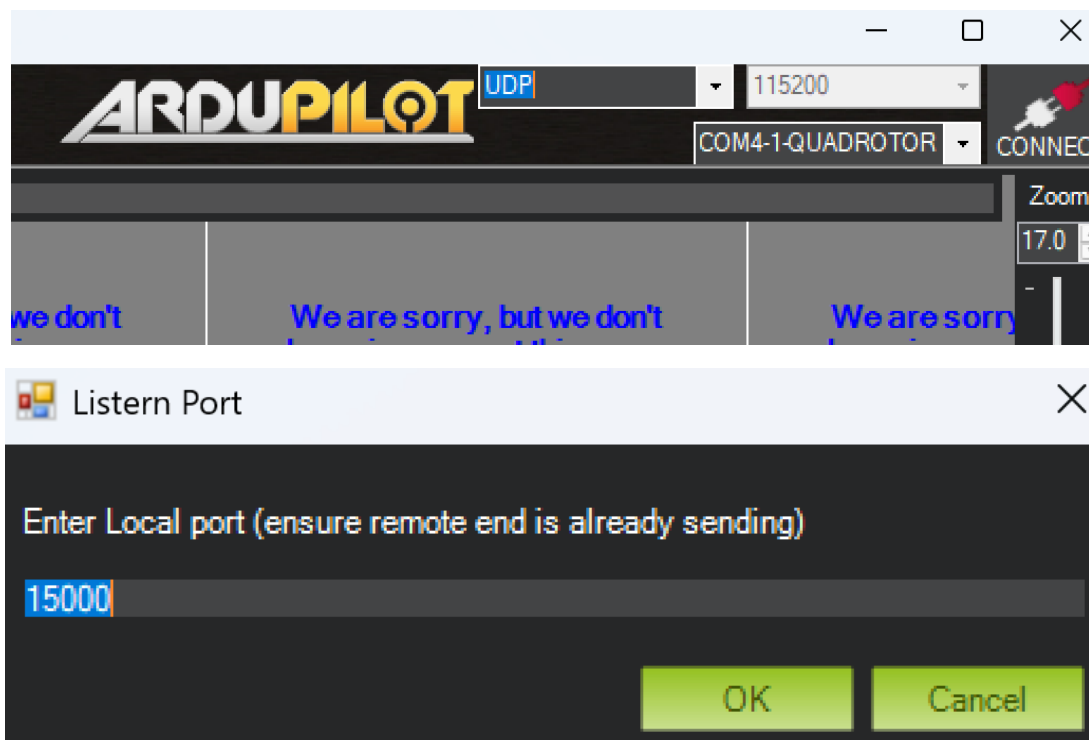
## Ground Station Software Connection Method

### Download the software package

Currently, the ground station only supports the Windows version of QGroundControl and Mission Planner. On a Windows PC with the ground station installed, extract the “TideLink-tel801-4g-intl-图传控制模块-行业无人机-软件包-V20250815.zip” software package.

## Ground Station Access (QGC, MP)

1. Enter “**UAVGCS\_RTCReconnect**” folder, edit **conf.yaml** file
2. Edit the number after "sn:" to the IMEI number of the 4G module's metal shell on the back of the Puffin module. For example, sn: 867652073902932.
3. Do not change “rtcServer: 54.176.172.85” in the file, as this is the IP Address of the test signaling server provided by Puffin.
4. Start “**UAVGCS\_RTCReconnect.cmd**”
5. On the PC interface, the CMD program running window and running log will appear. If you see "Connected" within the first 10 lines of the log, it indicates that the serial port signal of the flight controller has been pass-through transmitted to this PC, and the Puffin module has established a control data connection channel.
6. Run the customer's existing ground station, **select UDP 115200 connection, and choose local port 15000**,



7. After a few seconds, it can be seen that the ground station can display the telemetry information of the flight control system in real time.

## Image Access

1. Enter the “**tel801-行业无人机图像通道-webapp-v2024**” folder



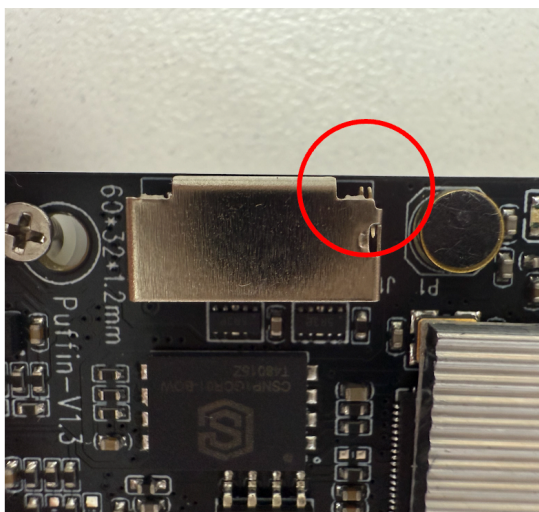
2. Run on a PC browser (Chrome is recommended) “**TideLink\_RTC\_Video.html**”, and fill in the same IMEI number at Channel ID as shown in the figure below. **Note to keep the.AV suffix.**
3. Click the Connect button, and within a few seconds, you will see the player's spinning play logo, and immediately the video from the RTSP camera connected to the Puffin module will be displayed.

|                                                                                  |                                                |
|----------------------------------------------------------------------------------|------------------------------------------------|
| Channel ID                                                                       | <input type="text" value="1481170f0143c.av"/>  |
| Client ID                                                                        | <input type="text" value="88590966223813388"/> |
| Video Format                                                                     | <input type="text" value="none"/>              |
| <input type="button" value="Connect"/> <input type="button" value="Disconnect"/> |                                                |

## Debug

### SD card is not inserted securely

Many unsuccessful cases are due to the SD card not being fully inserted into the card socket, causing the entire system to fail to boot up. Observe the reed on the upper right corner of the SD card socket. If the card is fully inserted, the reed will be in a closed state. If you confirm that the card has been inserted but the reed is not closed, you can use tweezers to pinch it.



无卡状态：簧片打开

No card state: Reed open  
Reed closed (short circuit)



有卡状态：簧片闭合（短路）

Card present state: